SDC

Hackathon · Technical

Technical Tools Declaration

Lumina Nusantara · Domain 1 — The Aiversecity Future-Ready Workforce Domain

SDG Track: SDG 4 — Quality Education (sub-target 4.4, with 8.5)

Team Leader: DU YULIN · Xiamen University Malaysia (XMUM)

Live prototype: https://sdc2026-lumina-nusantara-65cnugvuxdle5dgp9ky3fs.streamlit.app

Code repository: https://github.com/douyinlin/sdc2026-lumina-nusantara

Submission deadline: 25 October 2026, 07:59:59 UTC+8

> Technical Tools Declaration. Revised unsigned copy; recorded the actual status. Leader review and signature remain required.

1. TEAM AND PROJECT OVERVIEW

Team / project	Lumina Nusantara / SkillRadar MY
Leader	DU YULIN
Contact	CYS2609009@xmu.edu.my / +60 11-1798 1756
University	Xiamen University Malaysia (XMUM)
Live prototype	https://sdc2026-lumina-nusantara-65cnugvuxdle5dgp9ky3fs.streamlit.app
Public repository	https://github.com/douyinlin/sdc2026-lumina-nusantara

2. ARCHITECTURE & TECHNOLOGY STACK

2.1 Architecture

Streamlit collects a synthetic profile and calls local Python modules for keyword extraction, five-component matching, learning pathways and SDG interpretation. JSON files supply roles, courses and dated trend snapshots. Session state holds the current interaction. A validated manual import updates trend snapshots after human review. There is no scheduled ingestion, relational database, external LLM or GPU inference. Scenario and scoring coefficients are prototype heuristics, not measured hiring probabilities.

2.2 Runtime libraries

Validated local environment: Python 3.12 (PSF). Direct requirements: Streamlit 1.66.0 (Apache 2.0), pandas 3.0.6 (BSD 3-Clause), NumPy 2.5.4 (BSD 3-Clause), PyArrow 26.0.0 (Apache 2.0), Matplotlib 3.11.2 (Matplotlib license). Development/documentation utilities include unittest, mypy, Playwright, ReportLab, pypdf and pypdfium2. Exact ancillary versions and provider terms require final inventory confirmation. These utilities are not inference models.

3. APIs & CLOUD SERVICES

3.1 External APIs: No external model API is called by the product. Public report pages are source references, not continuously consumed API feeds. Firecrawl was used during development/review for web retrieval, not by the runtime app.

3.2 Services: Streamlit Community Cloud hosts the demo; GitHub hosts public code; Replit is evidenced as a development workspace. Current account plans, deployment region and deployed commit must be confirmed from account settings; they are not inferred from URLs. Replit project: https://replit.com/@duyulin18/SkillRadar-MY-Streamlit-app. Unauthenticated access on 23 October returned Page not found / owner invitation required.

4. AI TOOLS, MODELS & DATASETS

4.1 Runtime models: None. Skill extraction and matching use team-authored rules. The catalogue is not a verified complete O*NET/ESCO mapping.

4.2 Data: DOSM Q1/Q2 2026 occupation observations; WEF Future of Jobs 2025 global forecast; JobStreet/Randstad 2025 reported skill signals; role-level Malaysian references including PIKOM; team-curated role/skill and SDG catalogues; 36 course-link records; synthetic profiles and sample CV. See the source inventory appendix for URLs. Course costs/durations and source-specific reuse terms require verification. Team-authored code is Apache 2.0; this does not relicense third-party reports. No real personal CV data is intended for use.

4.3 Development AI: OpenAI Codex was used extensively for code review, implementation, debugging, tests and documentation. ChatGPT assisted project planning and written guidance. DeepSeek-based assistant and GitHub Copilot were declared in the earlier team document (scaffolding/documentation and inline completion

respectively); the team must confirm their actual usage and model versions. Product subscription names are not exact model identifiers. No unsupported model version is asserted.

Review statement: Automated checks and local browser checks are recorded in team/DEVELOPMENT_LOG.md. These do not prove all facts, courses or AI output were manually verified. Team review of code, data, assertions and final documents remains required. No claim is made that every change received independent pull-request review.

5. SPONSOR-PROVIDED RESOURCES

Replit was used as evidenced by the appended historical workspace screenshots. Those screenshots do not establish that the latest GitHub commit runs there. Restore judge access to project history or provide an acceptable live Replit deployment URL before submission. No RunPod endpoint or GPU is used by the current app. Replit is replaceable as a Python development environment; the current public demo runs on Streamlit Community Cloud. Account plan and sponsor credit usage require owner confirmation.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1 Privacy: Use only synthetic CVs and public or synthetic data. Uploaded text is parsed in memory; the application does not intentionally persist CVs in a database. Hosting provider/session behavior remains subject to provider terms. The app requires no inference API secret. Repository secret review and provider settings remain team responsibilities.

6.2 Limitations: Source lag, selection bias and incomplete course coverage affect recommendations. Keyword parsing supports plain text and limited text PDFs, not OCR or semantic understanding. Fixed catalogues can still contain incorrect roles, links or metadata. Prices and durations are not fully verified. Scenario coefficients and skill-to-SDG links are team interpretations. Fairness and user impact have not been established. The tool assists career exploration and must not make hiring decisions.

7. COMPLIANCE & READINESS

Before signing: confirm actual tools and versions, source terms, team review, workshop attendance, no exposed credentials, committee collaborator access, Replit proof accessibility and the deployed user flow. Add the YouTube video separately. Unresolved items are tracked in submission/OWNER_ACTIONS.md; they are not marked completed in this revision.

8. DECLARATION

The leader must review and sign the final accurate declaration. The old signature and future date have not been reused on revised content.

Leader signature (after review): _____

Name: DU YULIN

Actual signing date: _____

Appendix A: Public source and resource inventory

URLs recorded in the current JSON catalogue. Inclusion does not certify accuracy, licensing, fees or durations; these remain subject to team review.

https://academy.hubspot.com/courses/content-marketing

https://corporatefinanceinstitute.com/collections/

https://developers.google.com/machine-learning/crash-course

https://developers.google.com/tech-writing

https://explore.skillbuilder.aws/learn

https://hrsea.economictimes.indiatimes.com/news/industry/hitting-the-wall-70-of-malaysian-graduates-stuck-in-low-skilled-jobs-experts-warn/121397820

https://pikom.org.my/Reports-Job_Market/PIKOM_EDJMO_2025.pdf

https://skillshop.exceedlms.com/student/catalog

https://theedgemalaysia.com/node/791561

https://trailhead.salesforce.com/

https://wiki.freecad.org/Manual:Introduction

https://www.accountingcoach.com/bookkeeping/explanation

https://www.blackmagicdesign.com/products/davinciresolve/training

https://www.canva.com/designschool/

https://www.coursera.org/learn/human-resources

https://www.coursera.org/learn/negotiation-skills

https://www.coursera.org/learn/psychological-first-aid

https://www.coursera.org/learn/public-speaking

https://www.coursera.org/learn/research-methods

https://www.coursera.org/learn/teaching

https://www.coursera.org/learn/uva-darden-digital-product-management

https://www.coursera.org/professional-certificates/google-data-analytics

https://www.coursera.org/professional-certificates/google-digital-marketing-ecommerce

https://www.coursera.org/professional-certificates/google-it-support

https://www.coursera.org/professional-certificates/google-project-management

https://www.coursera.org/professional-certificates/google-ux-design

https://www.coursera.org/specializations/excel

https://www.coursera.org/specializations/intro-cyber-security

https://www.coursera.org/specializations/machine-learning-introduction

https://www.coursera.org/specializations/python

https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/

https://www.dosm.gov.my

https://www.dosm.gov.my/portal-main/release-content/big-data-analytics-job-market-insights-and-my-job-profile-first-q12025

https://www.dosm.gov.my/portal-main/release-content/big-data-analytics-job-market-insights-and-my-job-profile-second-quarter-of-

https://www.dosm.gov.my/portal-main/release-content/big-data-analytics-job-market-insights-q12026

https://www.dosm.gov.my/portal-main/release-content/big-data-analytics-job-market-insights-q22026

https://www.edx.org/certificates/professional-certificate/ritx-communication-skills

https://www.excel-easy.com/

https://www.freecodecamp.org/learn/2022/responsive-web-design/

https://www.freecodecamp.org/learn/relational-database/

https://www.kaggle.com/learn/intro-to-machine-learning

https://www.kaggle.com/learn/python

https://www.khanacademy.org/math/statistics-probability

https://www.netacad.com/courses/cybersecurity/introduction-cybersecurity

https://www.weforum.org/publications/the-future-of-jobs-report-2025/

https://www.weforum.org/publications/the-future-of-jobs-report-2025/3-skills-outlook

Appendix B: Historical Replit evidence

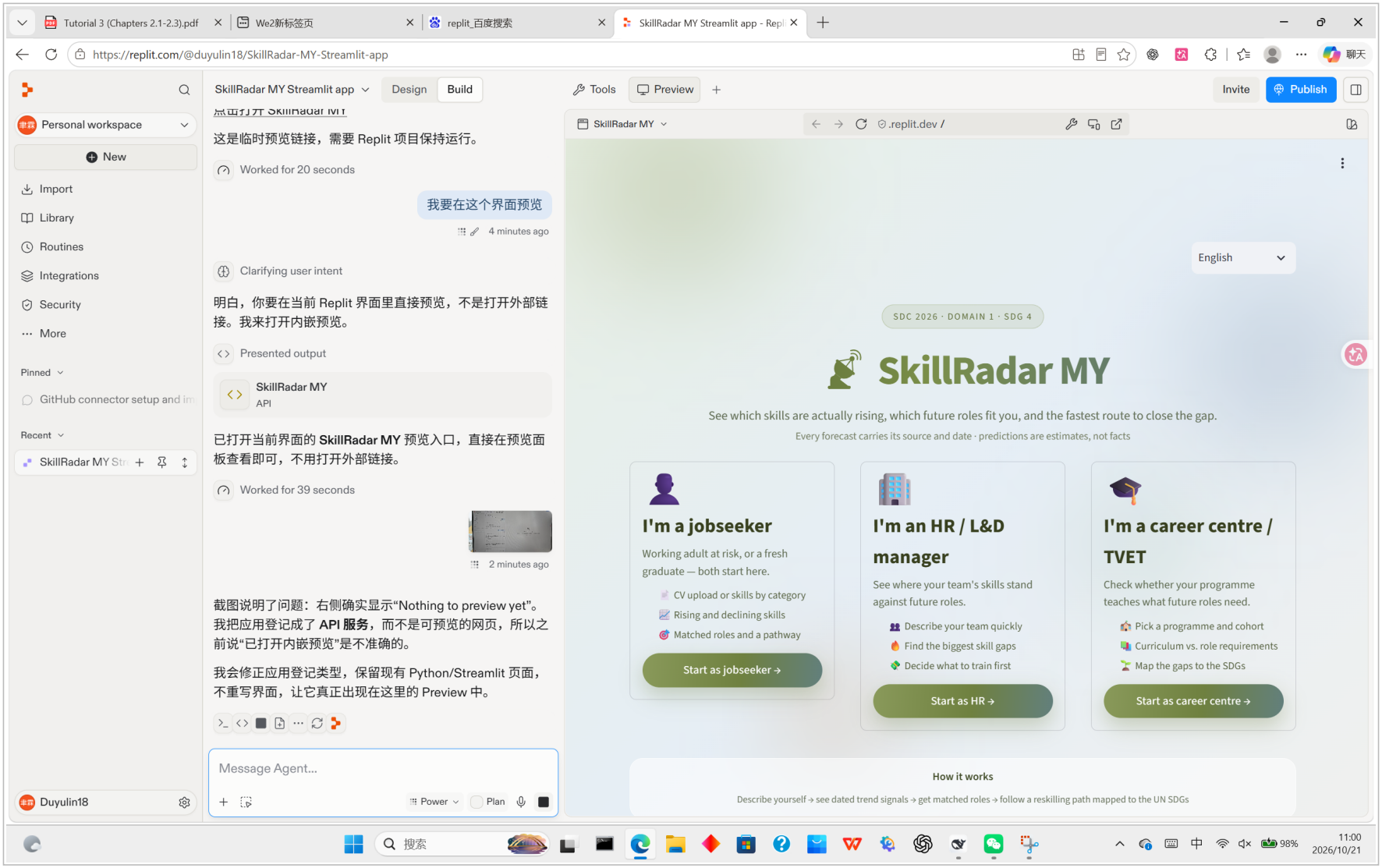
The following three pages are preserved historical screenshots. Their original captions refer to the version captured at that time, not the latest GitHub commit. They do not establish current judge access or current deployment equivalence. Project: https://replit.com/@duyulin18/SkillRadar-MY-Streamlit-app. Owner must restore access and verify the current version before signing.

Replit Proof of Use

SDC Hackathon 2026 · Domain 1 · Team **Lumina Nusantara** · Leader **DU YULIN** · Project **SkillRadar MY**
Replit account: **@duyulin18** · Repl: **SkillRadar-MY-Streamlit-app**

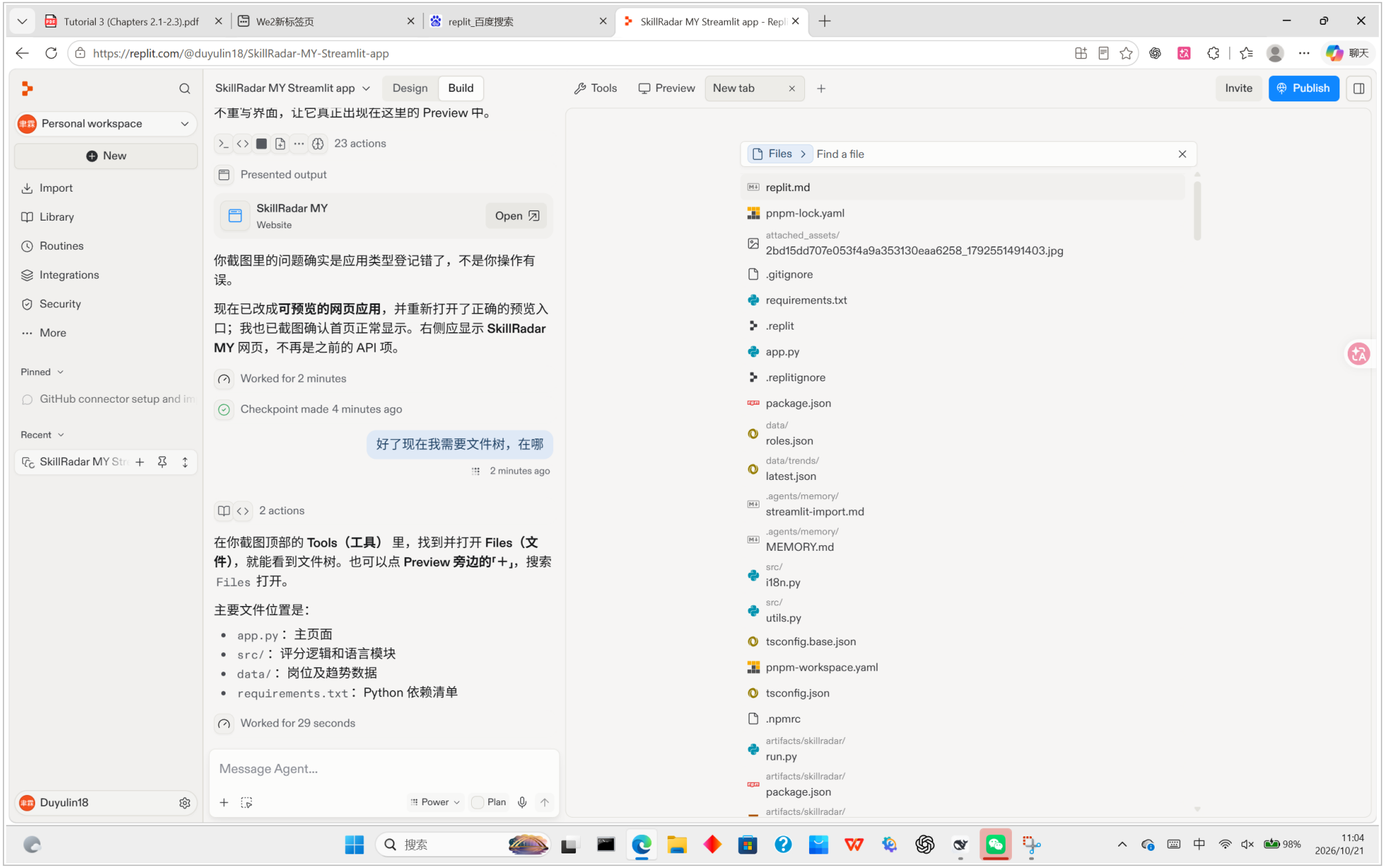
1 — The application running inside Replit

SkillRadar MY is open in the Replit workspace and serving its three entry points (jobseeker, HR / L&D;, career centre / TVET). The address bar shows the project URL replit.com/@duyulin18/SkillRadar-MY-Streamlit-app.



2 — The project files inside Replit

The complete source is present in the Repl: app.py (Streamlit entry point), src/i18n.py and src/utils.py (scoring engine), data/roles.json and data/trends/latest.json, plus requirements.txt. This is the code that runs the public demo.



3 — Version history inside Replit

Work on the prototype is tracked by Replit's own version control, with a commit history from the first scaffold through to the current implementation.

